

are heated. This voltage is proportional to temperature. Characteristics include wide temperature range (up to +2300°C), low cost, very low output voltage (about 40µV per °C for K type), reasonable linearity and moderately-complex signal conditioning. Signal conditioning for thermocouples requires a look-up table or algorithm correction.

Thermocouples are typically selected because of their low cost, high temperature limits, wide temperature ranges and durable nature. Applications for thermocouples range from industrial such as power generation, pharmaceutical and biotech to scientific and OEM applications. These are also found in everyday appliances like stoves, furnaces and toasters.

Measuring temperature with a thermocouple can be challenging because output signals are typically in milli-volt range, and generally thermocouples have very low temperature-to-voltage sensitivity. Therefore sources of errors related to thermocouple measurements that can impact measurement accuracy should be carefully considered. These are cold-junction compensation (CJC) errors, offset and gain errors, noise errors and thermocouple errors. Signal conditioning for a thermocouple involves CJC, amplification, isolation and filtering to remove these errors.

Let us walk through some important sources of errors and signal conditioning requirements in thermocouples:

Amplification. As output signals from thermocouples are typically in milli-volt range, these should be amplified. Gain is chosen according to the input limits of the ADC. Low-level voltages amplified near the signal source or measurement point typically improve the system's noise performance.

CJC. It is important to consider CJC error, which is one of the largest contributors to the overall

accuracy of measurements using a thermocouple. Connecting a thermocouple to a measurement device, in fact, creates three dissimilar metal junctions in a circuit: thermocouple junction (or hot junction) and junctions between each lead and measurement device (or cold junctions).

These cold junctions produce their own thermoelectric voltages proportional to the temperature at the device terminals. The technique used to remove this unwanted effect is called CJC, which uses another temperature sensor (usually a thermistor) to measure the cold-junction temperature, and uses this value to eliminate the parasitic thermocouple's effects.

Noise errors. As thermocouple output signals are very low, typically between -10mV and 80mV, these are susceptible to noise introduced either by the external environment or by the measurement device. Thermocouple data acquisition (DAQ) systems generally use low-pass filters (LPFs) to eliminate high-frequency noise.

Common-mode noise and ground loops. Another source of noise arises when thermocouples are connected/soldered to a conductive material such as steel or submerged in conductive liquids such as water, which makes thermocouples susceptible to common-mode noise and ground loops. Isolation is required to prevent ground loops and improve the rejection of common-mode noise.

Thermocouple error. Another source of error is temperature gradients across the thermocouple wire due to impurities in the metals.

Thermistors

Thermistors are temperature-dependent resistors usually made from metal-oxide ceramics or polymers, whose resistance changes substantially with temperature, more so than standard resistors. Most of these are negative temperature coef-

ficient (NTC) thermistors with their resistance dropping with temperature as opposed to standard resistors.

With a typical range of -100°C to +150°C, these are normally used for over-temperature shutdown purposes, monitoring the temperature of battery packs while charging, temperature of coolants and oil temperatures inside the engine to provide data to the engine control unit (ECU), and for temperature sensing in freezers and incubators.

Although thermistors are inexpensive and come in small packages, these are not as accurate as some of the other temperature-sensor solutions. These typically achieve greater precision within a limited temperature range. These are also non-linear, and their non-linearity can be addressed by software or by circuitry.

Of all thermistor disadvantages, self-heating is an important consideration. As discussed earlier, thermistors being resistive devices need an excitation current to read the voltage across their terminals, which results in self-heating. To minimise self-heat error to a low enough value, proper care must be taken to limit the sensing current.

RTDs

Platinum RTD is one of the most accurate sensors available for measuring temperatures within -200°C to +850°C range, capable of achieving calibrated accuracy of $\pm 0.02^\circ\text{C}$ or better. RTDs are resistive devices (often made from platinum wire) whose resistance varies with temperature. A 100-ohm platinum RTD, commonly called Pt100, has a typical resistance of 100 ohms at 0°C. Other characteristics include reasonable linearity and the need for signal conditioning. Cost of RTDs can be high, and these are available in probes, surface-mount packages and with bare leads.

RTDs are slowly replacing thermocouples in many industrial applications that work below 600°C due to their higher accuracy and repeatability.